

Instructor: Date:

Learner: Regional Branch:

Program: Center:

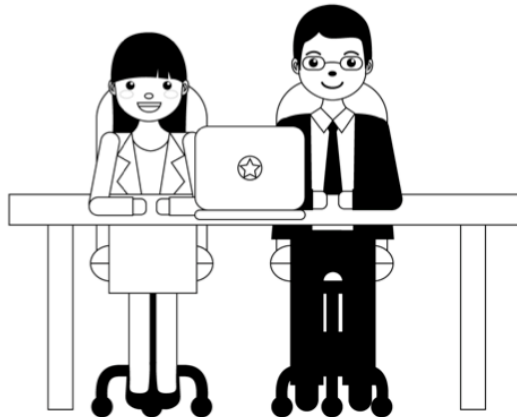
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Let's Start!

Software Engineering vs. Computer Science



Learning Outcome:

Understands systems theories, business processes and basic concepts on IT.
Create a map of processes that allows identifying areas in an information system, using IT tools to generate reports based on the company's needs.

Ask your partner:

- Do you prefer to apply engineering or scientific methods to software studies? Why, or why not?
- What do you think the advantages are for a software developer who knows about computer science?

Can Do Statements:

- I can talk about basic differences between software engineering and computer science.
- I can talk about simple differences between software engineering and system engineering.
- I can list the main challenges when facing software engineering.



Let's Explore!

Software Engineering in Context!



Instructions: Read the text below.

Oscar: Hi, Anna! What are you doing? You look worried.

Anna: Yes. I'm a bit confused. I need to decide what to study.

Oscar: What options do you have?

Anna: Software engineering, computer science, or systems engineering. But, I'm not sure about the differences between software engineering and computer science. Can you help me, please?

Oscar: Yes! Computer science is a scientific and useful approach for computer development. It isn't involved with software creation, it is more related to the research and extent of scientific knowledge.

Anna: I see. And what about software engineering?

Oscar: Well, software engineering is not an approach, but a discipline in which engineers develop different types of software. They do it with the help of principles of engineering and mechanics.

Anna: I see. Now, what can you say about systems engineering?

Oscar: Okay. Systems engineering is not an approach, it is an interdisciplinary subject of engineering. It dedicates efforts to the design and management of complex engineering systems. It doesn't have to do with the manufacturing of new products, but with the solution of problems associated with the physical systems. While software engineering controls software systems, systems engineering helps to develop advanced systems. For example, the Apollo program is a systems engineering development.

Anna: I can already design basic applications and programs, so I think software engineering is the best option for me.

Oscar: It isn't a bad decision. But remember that it has many challenges.

Anna: Really? Like what?

Oscar: Well, software techniques do not adapt to heterogeneous platforms, so the execution of programs in different environments is quite difficult. Software engineers need to develop techniques to deliver software more quickly, and better techniques for the demonstration of reliable and functional software.

Anna: I think that's a difficulty I can work with!





Let's Practice!

I'm not sure about the difference between computer science and software engineering.

Instructions: Read the dialogue. Underline all the mistakes in the paragraph. Correct the mistakes. Practice the conversation with your partner.

Ann: Sorry, the difference between computer science and software engineering am not clear to me.

Oscar: Well, computer science are not concerned with software creation. Software engineering not is an approach, it is a software production discipline.



Let's Practice!

Software and system engineering are totally different!

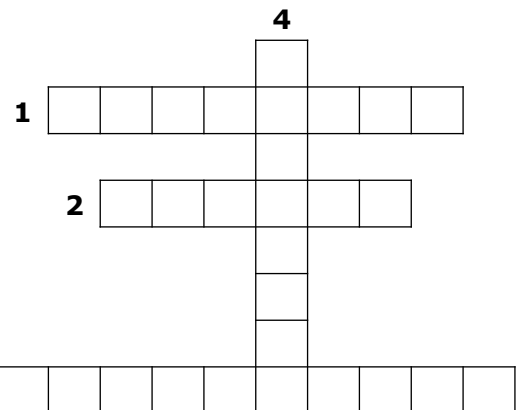
Instructions: Read the clues. Complete the crossword.

Across

1. Software engineering is a discipline; it is not an _____.
2. _____ engineering is not interested in manufacturing new products.
3. System _____ is not involved with the creation discipline, it develops advance systems.

Down

4. _____ and system engineering are totally different!



**Extension Activity**

Comparing software engineering, system engineering and computer science.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about software and system engineering and computer science.
- Describe what each approach or field is about.
- In a comparative chart, write what the importance of each one is.
- Share your chart with your class and decide on more descriptions for them.

Self-Assessment**Yes****No**

- I find information about software and system engineering and computer science.
- I describe what software, system engineering and computer science are.
- I understand the importance of software and system engineering and computer science.

☐ ☐☐ ☐☐ ☐**Target Words**

- **mechanics** (noun) science that study the design, construction, and operating machines
e.g. They do it with the help of principles of engineering and mechanics.
- **execution** (noun) the process of causing a computer to use a program or follow an instruction
e.g. Well, software techniques do not adapt to heterogeneous platforms, so the execution of programs in different environments is quite difficult.
- **platform** (noun) technology of the hardware and software of a computer that defines how a computer operates
e.g. Well, software techniques do not adapt to heterogeneous platforms, so the execution of programs in different environments is quite difficult.
- **approach** (noun) method used for dealing with a specific subject
e.g. Computer science is a scientific and useful approach for computer development.
- **heterogeneous** (adjective) integrated of particular and different kinds, combined
e.g. Well, software techniques do not adapt to heterogeneous platforms, so the execution of programs in different environments is quite difficult.
- **develop** (verb) to elaborate
e.g. Well, software engineering is not an approach, but a discipline in which engineers develop different types of software.



Learning Points

Simple Present Tense - Verb to Be Negative Form

The verb **to be** in the **simple present** tense is used to talk about a present or general state or the characteristics or location of something.

The verb **to be** in the **simple present** tense is used to talk about a present or general state or the characteristics or location of something. The **negative form** of the verb **to be** is created with the adverb **not**:

*It **is not** a science program.*

*She **is not** a software engineer.*

Take a look at the following chart to learn how the **negative** is formed with each personal pronoun:

To be - Negative Form	
Negative Form	Contracted Negative Form
I am not	I'm not
You are not	You aren't
He is not	He isn't
She is not	She isn't
It is not	It isn't
We are not	We aren't
They are not	They aren't

Examples

- *But, I'm not sure about the differences between software engineering and computer science.*
- *It isn't involved with software creation, it is more related to the research and extent of scientific knowledge.*
- *It isn't a bad decision.*





Learning Points

Software Engineering VS Computer Science and System Engineering

There are three different disciplines devoted to the computing process and everything that integrates it. Among these renowned disciplines you will find: **software engineering, computer science** and **systems engineering**. The functioning of computing programs, as well as hardware and software processes are common to all these disciplines. Let's see the differences and particularities of each one.

- **Software Engineering:** a discipline that applies engineering principles to the design, creation, and implementation of software.
- **Computer Science:** a scientific and useful approach to process information and make calculations for communication purposes.
- **System Engineering:** an interdisciplinary area of engineering devoted to the design and management of complex engineering systems.

Software Engineering	Computer Science	System Engineering
Software engineers work with proven knowledge to create software designs that work.	Computer scientists research and extend scientific knowledge. They also prove the accuracy of algorithms.	System engineers analyze and use interdisciplinary knowledge to produce physical systems.
It produces software in an economic and feasible way.	They test theories to patent reliable computing principles.	Focused on functionality to solve problems and create physical systems with quality outputs with minimum cost and time.
Common inventions: -Network distribution software for medical, industrial, military, communications, aerospace, business, scientific, and general computing applications.	Common inventions: -Algorithmic trading for financial market. -A formal definition of computation Modern digital computer. -Artificial intelligence -Simulation of various processes, including computational fluid dynamics, physical, electrical, and electronic systems and circuits.	Common inventions: -The Apollo program -Systems engineering techniques are used in complex projects: spacecraft design, computer chip design, robotics, software integration, and bridge building.